

Multi-Domain Recession Forecasting and Economic Regime Identification

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Introduction

Predicting the trajectory of a modern economy is a highly complex challenge that requires balancing historical patterns with the nuances of real-time market shifts. Traditional economic forecasting models often rely on single, domain-specific indicators such as housing statistics, GDP, and the yield curve. The logic underlying these models treat relationships as linear and are often threshold based (e.g. “If x drops below y, a recession is coming”). The binary logic employed in these models do not capture the complex interactions between other variables (e.g. an inverted curve might be “cancelled out” by high consumer spending”).

A similar approach is taken when categorizing economic “states”. Traditional economics classify an economy into two buckets, Expansion and Recession. The grouping of an economic period is based on the definitions set by the National Bureau of Economic Research (NBER) and are often lagging. This is to say, that you only know you are in a recession months after it has already started.

Our project aims to address both of the deficiencies in the approach to economic recession prediction and regime identification. We attempt to create a more robust early warning system for economic downturns and to identify latent economic regimes (beyond Expansion/Recession) in the months leading up to a recession.

From the beginning of the project, we understood that many economists, statisticians, and data scientists have attempted and continue to attempt creating models to accurately predict when the next recession will occur. It was likely we would not be able to “cure cancer” in the timeframe allotted to this project. Regardless, our team had a shared interest in economics and felt this project would help to develop our understanding of what economic factors affected countries the most?(sensitivity), the least?(resiliency), and how does each signal the overall health of an economy?

We make use of XGBoost (Extreme Gradient Boosting) in the primary supervised learning model for our prediction task. XGBoost (tree-based) was benchmarked against a simpler (probabilistic) logistic regression baseline and a specialized recurrent neural network, called Long Short-Term Memory (LSTM). Detecting latent economic regimes is accomplished through the use of Hidden Markov Models (HMM). The results were compared against K-Means clustering to determine whether the temporal structure modeled by HMMs provided superior regime segmentation compared to simple distance based clustering.

Related Work

Supervised

A study by Psimopoulos (2020), "Forecasting Economic Recessions Using Machine Learning: An Empirical Study in Six Countries," was conducted with a focus on predicting the pre-recessionary periods (last four quarters before a recession begins). Their goal is closely related to our supervised learning objective of predicting economic downturns across multiple international economies. Our project

improves upon Psimopoulo's methodology by integrating a multi-domain feature space that includes high frequency data, such as; Economic Policy Uncertainty (EPU) and market volatility (VIX), rather than the standard macroeconomic variables.

A Gogas et al. (2015) study, "Yield Curve and Recession Forecasting in a Machine Learning Framework," attempts to predict recession through machine learning classification. It specifically explores the use of the yield curve and its relationship between short term treasury bills and long term government bonds as leading indicators to predict future downturns. The study addresses the statistical challenges of identifying rate recessionary periods against periods of normal economic growth. Gogas limits its feature space to various interest rates for its modeling of the yield curve. As mentioned before, our project improves upon these simpler models through its inclusion of multi-domain data so that we can capture a much wider economic reality.

Unsupervised

A Chan-Lau and Wang (2020) study, "UnFEAR: Unsupervised Feature Extraction Clustering with an Application to Crisis Regimes Classification", shares the same underlying goal of our project (economic regime identification). In both instances, the task of regime identification is accomplished through the use of unsupervised machine learning methodologies. The formation and application of clusters is where our projects begin to diverge. Chan-Lau and Wang employed a boosted autoencoder designed to create time-invariant clusters, while we use Hidden Markov Models which explicitly calculates the sequential transition probabilities between economic states over time.

Data - Sources, Preprocessing, and Feature Engineering

Federal Reserve Economic Data (FRED)

FRED is a free online database maintained by the Federal Reserve Bank of St. Louis. It was established in the early 1990s to help better understand the Fed's policy decisions. It consists of economic data sourced from national, international, public, and private sources and can be accessed via API. A key component of our project is that our model be trained on multi-domain data. With that focus, we constructed our core dataset with variables from various categories covering Real Economy & Output, Financial Markets, Labor Market & Inflation, Financial Markets, Uncertainty & Leading Indicators, and Commodities. The target label, a binary signal showing which periods were recessionary, is provided by an indicator within FRED from the Organisation for Economic Co-operation and Development (OECD). Our raw data spans from 01/01/1970 to 12/01/2020, and covers Australia, Canada, France, Germany, Italy, Japan, South Korea, UK, USA. A total of 5508 records were returned in JSON format, converted into a pandas dataframe, and stored within a csv file¹.

Preprocessing

Several preprocessing steps were performed on the unstructured dataset to make it suitable for our models. Because we pulled from various sources and domains, our data contained daily, monthly, and quarterly series. To address this inconsistency, all data was harmonized to a standard monthly frequency. Quarterly metrics like GDP were upsampled using forward-filling and daily metrics were aggregated using monthly averages. To prevent data leakage within our model, indicators were shifted forward

¹ Dataset available in Github repo. (data/milestone_ii_dataset.csv)

according to their real world publication lags. This ensures that our model only uses data that would have historically been available at the time of prediction. Complete list of preprocessing steps can be found within notebooks Milestone_II_EDA.ipynb and Milestone_II_Preprocessing.ipynb

Feature Engineering

To address signal noise, non stationarity, and capture complex economic dynamics within our time series data, we engineered a wide range of features designed to improve our models' learning.

Feature	# of Features	Reason
Log Differences	40 Features	Remove trends and drift within data.
First Differences	12 Features	Remove trend; achieve I(0) integration
Amplitude	10 Features	Measure intensity / Energy of the signal
Yield Curve Features	4 Features	Capture term structure and macro-sentiment
Rolling Statistics	18 Features	Capture local trends and volatility

After engineering, our data set contained a total 237 features, 5508 Rows, with a class balance ratio of 3.6: 1 (Positive Class = 21.7%)

Supervised Learning

Methodology - Selection & Description

When determining the supervised learning technique to use within our model, we focused on finding algorithms best fit for time series analysis and that could do well with ragged edge data (not all countries began observing/recording economic indicators at the same time). Three algorithms were explored:

1. Logistic Regression
2. XGBoost
3. LSTM

Logistic regression: A probabilistic (linear) model is used often for binary classification tasks due to its high interpretability. It was chosen as a baseline in our project to compare against more complex algorithms so that we can objectively compare whether or not the non-linear capabilities of LSTM and XGboost actually provide a meaningful difference over a simpler (linear) approach. Because of the simple logic (assuming relationships to be linear) it can miss more complex relationships within our data. This deficiency is addressed through the creation of polynomial features. Refer back to the **Feature Engineering** section for more details.

XGBoost: Extreme Gradient Boosting is an advanced open source implementation of the gradient boosted trees algorithm. It works by sequentially building independent trees, with each new tree being specifically trained to correct residual errors made by the previous tree. It is able to natively capture complex, non-linear interactions between features, but treats rows as tabular observations (requiring the creation of

“lagged” features). A key reason behind our including this model was due to its ability to perform well on data with missing values. It does so by learning which direction (left tree branch or right tree branch) to send the missing data point so that loss is minimized.

LSTM: Long Term Short Memory is a specialized architecture of recurrent neural networks. It is designed for deep learning on sequential and time-series data. It is sequence based and differs from our prior models in how it processes data. LSTM works through data in sequential windows and contains an internal cell state and three gates (input, forget, output). LSTM’s strength is in its ability to capture long term temporal dependencies and momentum without the need for manual engineering, but it comes at the cost of interpretability and can be considered a “black box”. Similar to our motive behind using a basic logistic regression model, LSTM was included in our project to understand if a deeper and more computationally heavy learning approach could offer any significant benefit over a simpler tree based model.

Methodology - Evaluation

When working with a dataset as severely imbalanced as ours, optimizing a model on a single metric could lead to issues. We wanted to minimize false negatives so that the model isn’t constantly triggering alarms that a recession is impending, but we are also concerned with optimizing for false positives as we do not want our model to be overly cautious and fail to trigger when it becomes mathematically apparent that a recession is on its way. This led to our decision to use **Precision-Recall AUC** for our model performance measurement. PR-AUC allows us to see the tradeoff between recall and precision at every threshold and also gives us a score that can be interpreted to understand each model's ability to separate periods of recession from growth.

The data was split chronologically into 5 folds to prevent data leakage and any look-ahead bias. Within each fold, an 80/20 temporal split was applied to create our validation set. The 20% validation subset was also used as an evaluation set for early stopping in our XGBoost and LSTM models. Hyperparameters tuning for each of the models was done with the assistance of Optuna², an open source hyperparameter optimization framework designed specifically for machine learning. It works by running a series of trials using a Tree-structured Parzen Estimator (TPE). For each trial, the estimator suggests a new set of hyperparameters to evaluate based on the metric we are optimizing for. Tuning for our LSTM model was done over 20 trials due to the longer training times required. Two hundred trials were run for our other models. After tuning, the XGboost produced the best mean PR-AUC. This is likely due to its ability to handle missing values, as the logistic regression and LSTM models could only be evaluated on a limited subset of the data.

	Linear Regression	XGBoost	LSTM
PR-AUC	0.336 ± 0.190	0.478 ± 0.108	0.348 ± 0.160

² Details on Optuna: <https://optuna.org/>

Feature Importance

To understand which specific indicators were driving our XGBoost model, an analysis was conducted using a native measure of XGBoost called **Gain Importance**. Gain importance records how much the loss function decreased for every split in every tree. A feature's gain importance is the sum of these loss reductions across all splits where that feature was used, normalized to 1.

Our gain rankings reveal a strong concentration of importance in just a handful of features.

Key Findings

CLI (Composite Leading Indicator) and its deviation (**cli_dev**) together account for nearly half of all gain importance (~0.48). CLI is specifically designed to anticipate turning points in economic activity so its importance is economically intuitive.

CPI 12-month log-change (third rank)

illustrates the role of inflationary dynamics in recession forecasting. This is likely due to the tendency for central banks to raise rates in response to rising inflation, which can have a ripple effect that triggers a downturn.

Notably, 19 of 37 features show zero gain importance, meaning the model never split on them. This is common in gradient-boosted trees with early stopping when a small set of features provides sufficient predictive power, the model has no incentive to explore weaker signals. These zero-importance features were retained by the feature selection pipeline (permutation + LASSO consensus) but are effectively

unused by the tuned model. The error bars (standard deviations) suggest that the top features (CLI, cli_dev) are highly stable across folds, while mid-tier features like `10_yr_yld_diff_6mo` and `comp_consumer_conf_diff_12mo` show more variability.

Figure 1: XGBoost Feature Importance

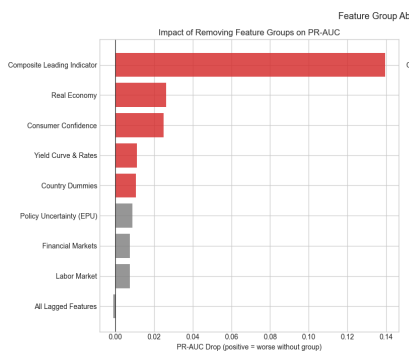
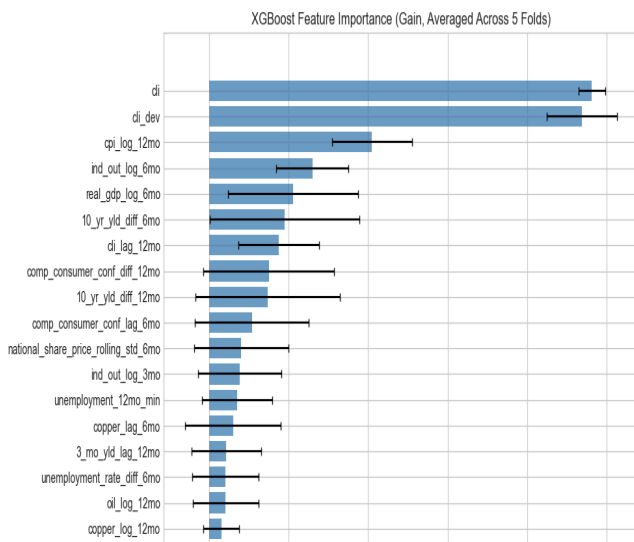


Figure 2: Feature Ablation

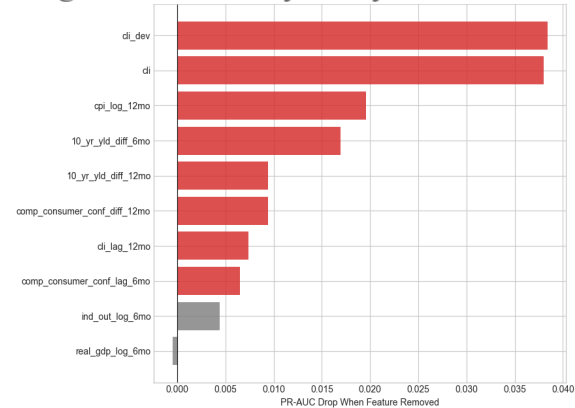
Country Dummies, Policy Uncertainty (EPU), Financial Markets, Labor Markets, All Lagged Features.

Key Findings:

Composite Leading Indicators dominate the model

The CLI group (3 features) accounts for a +0.14 PR-AUC drop when removed. That is nearly 30% of our models total performance. It is 5x larger than the next most important group (Real Economy). The CLI's dominance is explained by its design: it is itself a composite of multiple leading indicators (industrial production, stock prices, money supply, etc.) that are individually represented elsewhere in our feature set. Our model essentially learns to rely on this pre-aggregated signal rather than constructing it from raw components itself.

Figure 3: Sensitivity Analysis - Features



Feature Importance does not equal necessity.

The analysis revealed a critical distinction between feature importance and feature necessity within our model. The bottom two features are heavily used but can easily be replaced.

Feature	GI Rank	Ablation Rank	Status
cli	1st	2nd	Critical
cli_dev	2nd	1st	Critical
cpi_log_12mo	3rd	3rd	Important
ind_out_log_6mo	4th	9th	Replaceable
real_gdp_log_6mo	5th	10th	Replaceable

Lagged Features are unnecessary for XGBoost

All six lagged features can be removed without any performance loss. The difference based transformations are already encoded in the rate of change information that lags were designed to provide. XGBoost's tree structure can implicitly capture threshold-based temporal patterns from current values alone.

In terms of practical implications, the feature analysis and ablation revealed that our model can be simplified. Removing the 6 lagged features and the 12 financial market features would reduce the model from 37 to 19 features with only ~0.01 PR-AUC loss, a worthy tradeoff to increase interpretability.

Sensitivity Analysis

To test the robustness of our model's prediction when subjected to changes in inputs, thresholds, hyperparameters, and data noise we conducted a sensitivity analysis.

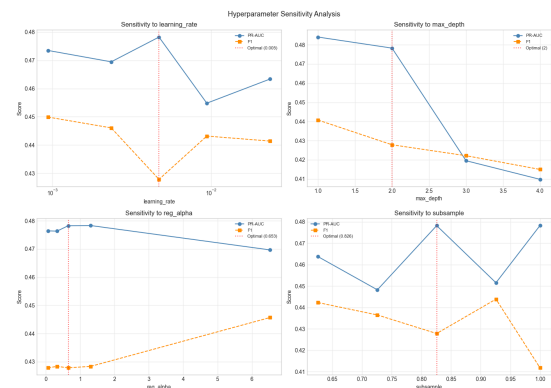
Key Findings

Feature Sensitivity: Our analysis tested the model's sensitivity to individual features by replacing the values of the top 10 features with their mean, as well as shifts of ± 1 and ± 2 standard deviations, while keeping all other features the same. We found that CLI was by far the most dominant driver and shifting it ± 2 STD resulted in a shift of up to 4.2% in predicted recession probability.

Hyperparameter Stability: Our usage of Optuna during tuning helped us find optimal hyperparameter values, but the search explored a limited set of candidates. To determine whether our hyperparameters sat at a stable plateau or a sharp peak in the performance space, we systematically varied four key settings; learning rate, max depth, reg alpha, and subsample. We found that our hyperparameter landscape was reasonably smooth and only max depth showed a strong gradient.

Overall our XGBoost Model was shown to be robust to data noise and hyperparameter perturbances, but exhibited fragility to threshold tinkering.

Figure 4: Sensitivity Analysis - Hyperparameters



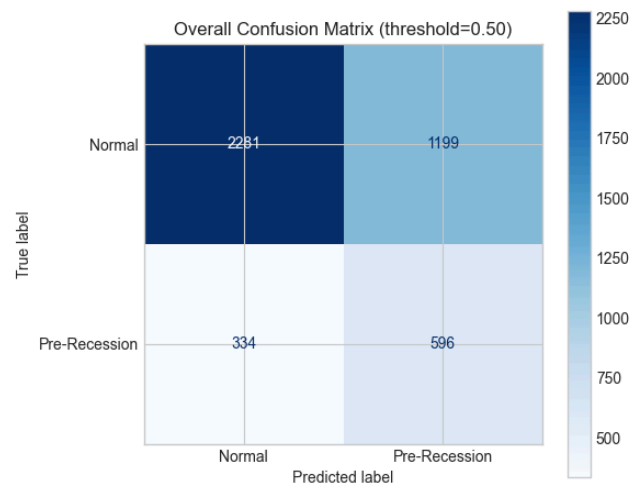
Failure Analysis

An examination of our models false positives and false negatives was conducted so that we could better understand where, when, and why the model fails.

Key Findings

Error Distribution: Our model produces roughly symmetrical error rates: FPR = 34.5% and FNR = 35.9%, meaning it mislabels about one-third of both negative and positive observations. From a practical standpoint, the high false positives count shows our model is triggering false alarms at a rate of one in every three months. This level of false-alarms would likely erode user trust in a production system. The almost equivalent false negative rate means that we are missing the same proportion of actual pre-recession months.

Figure 5: Failure Analysis - Matrix



Per-Country Failure Rates: When splitting rates by country, a clear geographic divide emerges in our model's performance.

- Best Detected Countries:
 - Italy (86.8% recall)
 - USA (77.4%)

- UK (76.1%)
- Worst Detected:
 - South Korea (45.5%)
 - Canada (48.2%)
 - Australia (55%)

Episode Detection: Of the 82 distinct pre-recession episodes across all of the countries, 19 were completely missed (0% detection rate) and 52 were well-detected (>50% detection rate). The majority of completely missed episodes occurred in the 1980's. This is likely due to their appearance in the first 1-2 validation folds where the expanding window training set is the smallest. Our model lacks sufficient training examples to generalize these early recessions.

Figure 6: Failure Analysis - Per Country



Lead Time Accuracy: For the 63 episodes that the model does detect (or partially detects), the lead time results are encouraging. Our analysis revealed that the model has an average detection delay of only 1.5 months into the pre-recession window, meaning that the model provides an actionable average lead time of 10.2 months before an actual recession started. In 38 out of 60 detected episodes, the model triggered a warning in the very first month of the pre-recession window.

The overall conclusion is that our model's failure mode is binary, it either catches the recession early or it misses it entirely.

Unsupervised Learning

Methodology - Selection

Two possible candidates were chosen to use in our model tasked with discovering latent macroeconomic regimes.

1. K-Means
2. Hidden Markov Models (HMM)

A subset of our data containing twenty features across a date range of 11/2001 to 12/2018 was used for our classification task. The features were selected using a three stage process (correlation removal, permutation importance, and LASSO L1).

Evaluation - K-Means

K-Means is a partitioning algorithm that groups unlabelled data into a predefined number of non-overlapping clusters. In our project it works by assigning every month of economic data to the nearest centroid. It then recalculates the center of each cluster based on the points assigned to it and

moves the centroid. Its strengths are that it is easy to implement, it is efficient, and the results are easy to interpret and visualize.

In the **K-means model**, four validation metrics were used to determine the best choice for the number of clusters from a range of two to ten. Three of the four metrics pointed to k=4 being our best operating point. After fitting, the model was grouped the dataset into the following categories.

Clust	Size	Share
0	491	58%
1	175	26.5%
2	1076	9.4%
3	112	6.0

After clustering, additional analysis was done to reveal the characteristics of each.

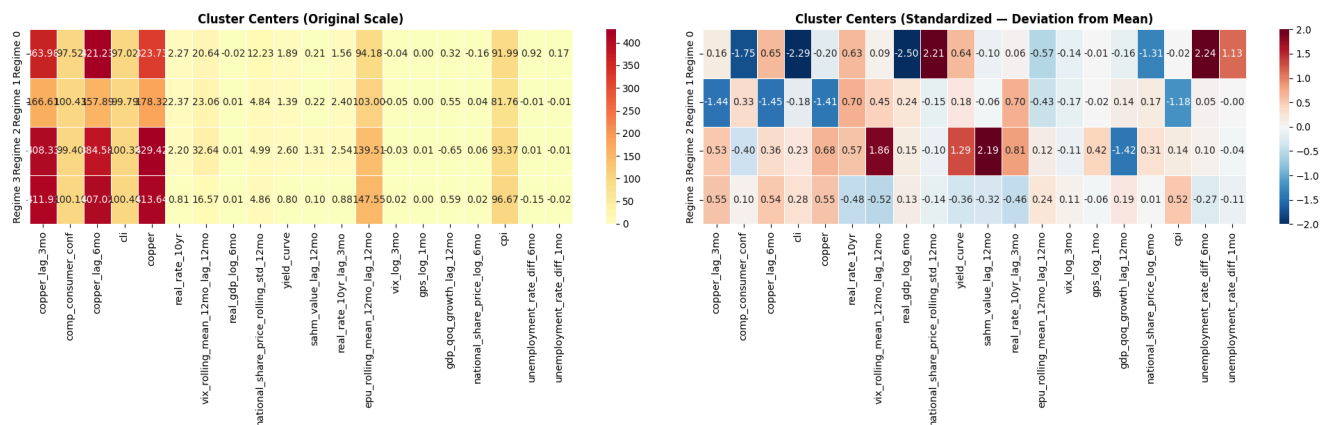


Figure 7: Clustering Analysis - K Means

Regime 0 (Acute Crisis - 6.0%): Characterized by sharply negative GDP growth, surging unemployment, extreme stock market volatility, and a below trend of CLI

Regime 1 (Low-Commodity / Early-Cycle - 26.5%): Characterized by very low copper prices and CPI, reflecting the early 2000s low commodity price environment.

Regime 2 (Post-Crisis Recovery 9.4%): Marked by high “memory” of recent stress (high lagged VIX and SAHM indicators), a steep yield curve typical of central bank easing, and improving current conditions.

Regime 3 (Broad Low-Volatility / Low-Rate State - 58.0%): The dominant “normal” state, reflecting low real interest rates, low volatility, and flat yield curves consistent with the post 2010 expansion environment.

Evaluation - HMM

Hidden Markov Models is a statistical model used to predict sequences by inferring hidden states from observable data. In our project the HMM assumes the economy is always in one of a few hidden states (regimes). It learns two main things, the typical shape of the economic data and the mathematical likelihood of moving from one state to another. It then calculates the most likely sequence of states that produced the historical data that was observed.

To determine the optimal number of hidden states to use in our **Hidden Markov Model** we evaluated state counts from 2 to 8 using the three metrics: Bayesian Information Criterion (BIC), Akaike Information Criterion (AIC), and Log-likelihood.

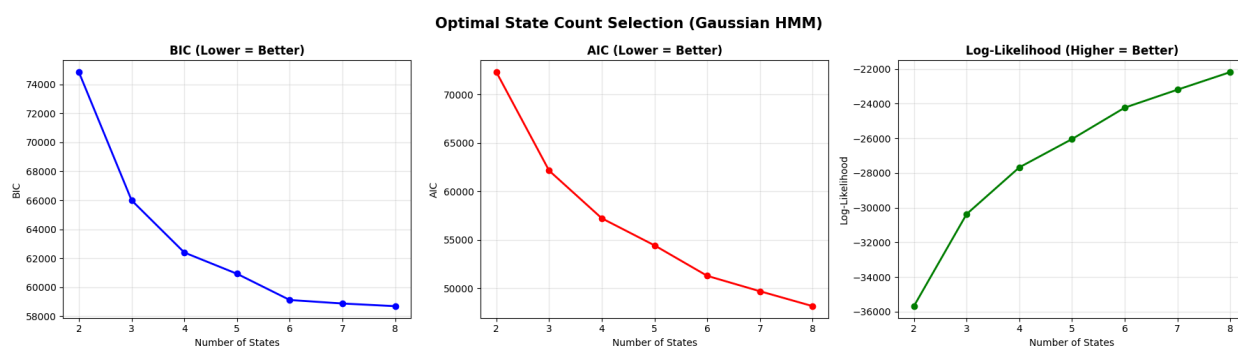


Figure 8: Optimal State Count Selection (HMM)

BIC evaluates the model fit while heavily penalizing unnecessary complexity (extra states). AIC works similarly to BIC but with a lighter penalty for complexity. The Log-likelihood measures how well the model fits the sequential data. All three metrics selected eight as the best number for hidden states. The takeaway from our results suggest that there is enough information in our data to justify more than a small number of regimes.

Further analysis of our 8 states revealed the following characteristics:

Regime 0 (Early-2000s Stable - 12.8%):

Characterized by low copper prices, moderate-high real rates, and low equity volatility. Reflects early-period conditions with low commodity prices and conventional monetary policy.

Regime 1 (Policy Uncertainty - 15.8%):

Characterized by high copper prices and the highest levels of Economic Policy Uncertainty (EPU). Consistent with periods of high commodity prices and contentious fiscal or monetary policy debates.

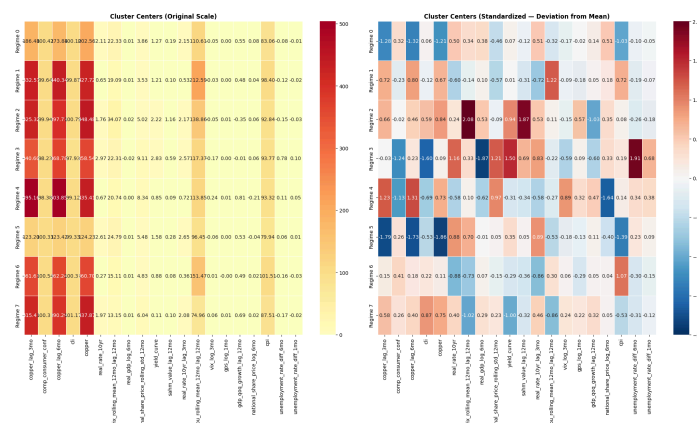


Figure 9: Clustering Analysis - HMM

Regime 2 (Post-Crisis Recovery - 8.7%): Marked by a high "memory" of stress through lingering unemployment and volatility indicators, combined with a steep yield curve typical of accommodative policy following a recession.

Regime 3 (Recessionary Stress - 6.7%): Characterized by negative GDP growth, accelerating unemployment, and high equity volatility. Reflects an active contraction phase paired with aggressive monetary policy responses.

Regime 4 (Short Transitional - 7.0%): A brief bridge state marked by near-zero growth, declining copper prices from peaks, and sharp equity declines. It has a short expected duration of only 6.8 months.

Regime 5 (Pre-Commodity-Boom - 11.9%): Characterized by the lowest copper prices and lowest CPI in the set. Reflects the early-period economic environment prior to the global commodity supercycle.

Regime 6 (Low-Rate Expansion - 21.9%): The dominant long-run regime, characterized by the lowest real rates and highest CPI. Reflects the post-crisis accommodative environment with a high 44.3-month expected duration.

Regime 7 (Mature Expansion - 15.1%): Marked by the highest CLI, lowest volatility, and lowest uncertainty. Reflects a broad-based expansion with high confidence, though a near-flat yield curve may signal late-cycle dynamics.

Item	K-Means	Hidden Markov Model (HMM)
Data used	Same (1854 rows, 20 clustering features, 9 countries, 2001-11 to 2018-12)	Same
Chosen number of regimes	k=4 (best silhouette)	8 states (best BIC/AIC in tested 2–8)
Silhouette	0.2403	0.0839
Calinski-Harabasz	330.3710	194.8295
Davies-Bouldin	1.5278 (lower is better)	2.4903
Stability (50 seeds, ARI mean)	0.9843 (highly stable)	0.5836 (more sensitive to init)
Regime size balance	Very imbalanced (largest cluster 58.0%)	More balanced (6.7% to 21.9%)
Temporal dynamics	No transition modeling	Yes; strong persistence (self-transition ~0.854 to 0.977)

While modeling regime evolution over time, HMM is more informative, but with weaker geometric clustering metrics and slower stability.

Sensitivity Analysis

We conducted a security analysis on our unsupervised Hidden Markov Model with its baseline configuration of 8-state HMM regimes are not an artifact of a single fit, we ran a sensitivity analysis covering random initialization (seed), feature shocks, hyperparameters, measurement noise, country leave-one-out, and time-based train/test splits.

Key findings

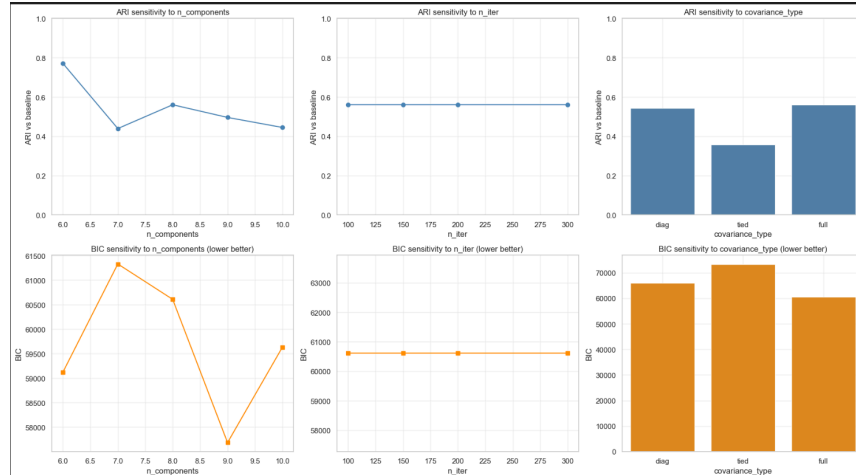
Feature Importance: The regime test reacts most strongly to “copper” variables(both current and lagged) and a handful of stress indicators such as VIX, SAHM, and yield-curve measures. Large shifts in these features can substantially re-map regime labels, indicating that several regimes are “defined” primarily along commodity/stress dimensions rather than evenly distributed across all features.

OAT feature sensitivity summary:

	feature	max_ari_drop	mean_l1_shift	mean_conf_change
2	copper	0.8155	1.2700	-0.0041
3	copper_lag_3mo	0.8093	1.0408	-0.0015
8	vix_rolling_mean_12mo_lag_12mo	0.5573	0.9261	-0.0069
6	sahm_value_lag_12mo	0.4449	0.7115	-0.0017
9	yield_curve	0.2217	0.2617	-0.0049

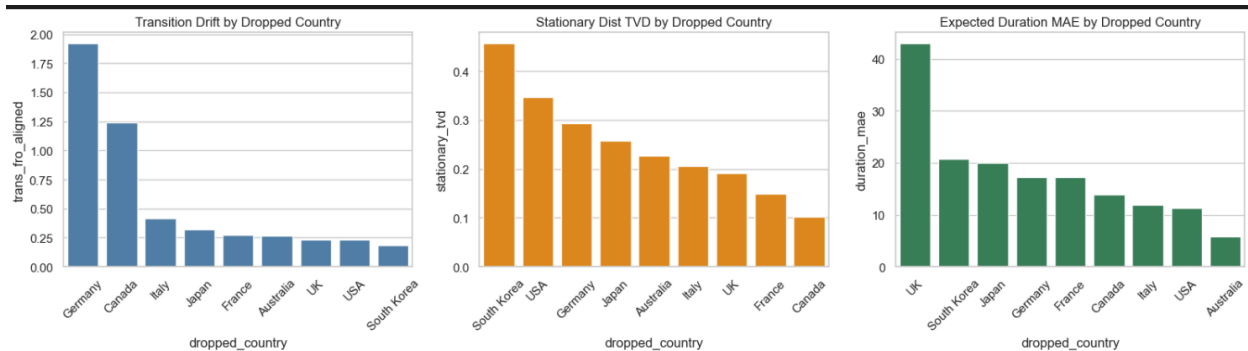
Hyperparameters: We tested how sensitive the HMM regimes are to key hyperparameters by changing one setting at a time and measuring both fit (Log-Likelihood, BIC) and regime consistency (ARI vs the 8-state baseline). The results show a clear fit vs. label-consistency tradeoff, a 9-state model optimizes statistical fit in our grid search, while a 6-state model most closely reproduces the baseline regime labeling.

- Number of states (**n_components**) is the dominant driver of regime definitions. Across the grid, it produced the largest ARI spread (0.331), meaning changing the state count leads to the biggest shifts in regime assignments. In terms of fit, the best-performing configuration by both Log-Likelihood and BIC was **n_components = 9** (best BIC = 57,683.8).
- Covariance structure (**covariance_type**) also affects results, with a meaningful ARI spread of 0.202, indicating that assumptions about within-state variance/correlation materially change regime boundaries.
- Iteration count (**n_iter**) has negligible impact after convergence. It showed 0.000 ARI spread, implying that once the model converges, running more iterations does not change the learned regimes.



Noise sensitivity: Small input perturbations ($\leq 5\%$ noise) barely affect assignments ($\text{ARI} \approx 0.99$), but larger corruption reduces stability (≈ 0.87 ARI at 20% noise).

Country jackknife (drop-one-country refits): Drop each country from training, refit, decode the full dataset, align states, and compare transition/stationary metrics. Dropping **Germany** causes the largest transition drift ($\text{trans_fro_aligned} = 1.919$), making it the most influential country for global transition dynamics. The Least transition impact is **South Korea(0.182)**.



Overall, the 8-state HMM gives very confident regime labels, but the exact regimes aren't perfectly stable. Changing the random seed can noticeably reshuffle states ($\text{ARI } 0.357\text{--}0.722$), and the results are especially sensitive to a few drivers—most notably **copper** (and its lag) and **lagged VIX**—where simple shocks can cause big label changes (max ARI drop ~ 0.816). The biggest differences come from **how many states** we choose and the **covariance structure**, not from running more iterations. Even after aligning labels, the transition behavior still shifts (mean drift 0.774), and time-split tests generalize poorly (worst OOS ARI 0.0089), so we report these regimes with uncertainty rather than as a single “final” truth.

Discussion

Supervised Learning:

We learned that the XGBoost was the best performer for our recession prediction task. Within that model, the Composite Leading Indicator (CLI) and its deviations were the backbone of the predictive signal. In our ablation analysis we found that removing just three of the CLI features caused a massive 29%

collapse in PR-AUC (dropping from 0.478 to 0.339). Our lagged features proved redundant in the XGBoost model as its tree architecture successfully learned the temporal dynamics and threshold patterns directly from contemporaneous values and diff-based transformations. Another interesting finding was how little predictive power all the features from our financial markets (VIX, oil, copper, and share prices) had. They yielded a very low average gain per feature (0.005) and could be dropped entirely without much impact on the model.

What surprised us the most was how diminished the role of the Yield Curve was within our model. From the beginning of our research we learned that the Yield Curve was one of the most celebrated recession predictors in macroeconomics. We assumed that it would prove to be important in our model, but it was actually assigned a secondary role. Removing all six of the Yield Curve features only caused a minor PR-AUC drop, likely because the CLI already incorporates interest rate signals.

Data scarcity was our biggest challenge in completing the supervised learning component of our project. Both the logistic regression and LSTM models required a completely NaN-free dataset. This forced us to drop a large portion of incomplete rows which severely reduced the training data for those models from 5,292 down to 1,854. If given more time we would dedicate it to finding new indicators that are less redundant and that have a lengthy history of being tracked by all of the G7 nations.

Unsupervised Learning

We learned that macroeconomic cycles are far more nuanced than simple binary expansion or recession labels. Within our K-Means model, we identified four distinct economic regimes: an acute crisis, a low-commodity era, a post-crisis recovery, and a broad low-volatility/low-rate state. Our Hidden Markov Model (HMM) expanded on this by finding an optimal temporal structure of 8 states. We also found that economic states are highly persistent, with the HMM showing self-transition probabilities ranging from 85.4% to 97.7%. When economies do transition, they follow sparse, identifiable pathways rather than switching randomly.

What surprised us the most was how the K-Means clustering naturally separated recessionary from non-recessionary periods, even without being trained on recession labels. Although the mapping did turn out to be highly asymmetric. Regime 0 heavily with recessions (79.5%), while Regime 2 (post-crisis recovery) had a surprisingly low recession rate (14.9%) despite the elevated lagged stress indicators.

We faced challenges with initialization sensitivity in our Hidden Markov Model. The Expectation-Maximization algorithm used for the HMM struggled with local optima in the high-dimensional feature space. After refitting the model 50 times it yielded an Adjusted Rand Index of just 0.584, showing significant instability and sensitivity to starting conditions.

If given more time we would try to construct a hybrid model for our unsupervised learning task. Ideally we would be able to combine the strong initialization stability of the K-Means approach (ARI of 0.984) with the sequential likelihood optimization of the HMM to create a framework that values both geometric compactness and temporal dynamics.

Ethical Considerations

Deploying the **supervised learning model** comes with ethical implications due to the potential effects it could have on macroeconomic policy and human livelihood. Our failure analysis that our XGBoost model

produced a high number of false positives, often clustering around periods of genuine economic stress that ultimately stopped short of a true recession. The ethical risk here is that we create a self-fulfilling prophecy. If our tool is relied on heavily by policymakers, corporations, or financial institutions and they react accordingly to a false positive by cutting investments, freezing hiring, or reducing their work force, those actions themselves can artificially induce the (false) recession that our model predicted. To address this we must explicitly frame our model as a supplemental decision support tool rather than an automated trigger to economic policy changes. An additional step would be to implement a “confirmation period” rule requiring sustained consecutive months above our threshold probability before the model signals an alarm. The false negative rate of our XGBoost carries its own set of potential issues. Our model completely missed 19 out of 82 recession episodes. If government agencies were to rely on our model for planning economic interventions, a missed episode would mean a failure to prepare stimulus packages, social safety nets, or unemployment reserves. The human cost of missed predictions would fall most heavily on the workers who are now impacted by sudden economic downturns. One solution towards addressing this would be to treat our decision threshold as an ethical level, instead of just a mathematical one. This would mean refining what the primary objective is for using the model. For example, if the goal is to protect vulnerable populations, we could intentionally lower the decision threshold to prioritize recall over precision. This would lead to more false alarms but also ensure that safety nets are funded and countries are always well prepared.

Our **unsupervised learning model’s** ethical implications can arise if our classifications are used to define macroeconomic reality and influence policy decisions. Troubles would arise if we were to present our clusters to policymakers as pure, unbiased discoveries of latent economic states. Our pipeline utilized upstream feature selection that was heavily informed by recession targets, which inherently creates a bias. Without disclosing that information, policymakers could believe these clusters to be completely objective and free of historical biases. That misunderstanding could lead them to over trusting the model to make completely neutral policy decisions. To avoid this issue we would need to be vigilant with transparency in our introduction and training of the model to its users. It would require explicitly documenting and presenting our findings as “target informed” regime discovery.

Statement Of Work

Matthew Zhou: Matthew was responsible for development, tuning, and evaluating our supervised and unsupervised learning pipelines. This included creating the baseline models for the XGBoost, Logistic Regression, LSTM, and K-Means. Matthew also suggested the topic for this project and played a large role in data mining, EDA, and in refining each notebook to make them easier to understand.

Dongxin Liang : Dongxin was responsible for the post tuning analysis of our models, this included conducting the ablation, failure, sensitivity, and feature analysis that was done on our XGBoost model and completed the sensitivity analysis on our HMM model. Dongxin acted as the primary liaison to our MADS team mentor, and was responsible for scheduling our check-ins when needed.

Christopher Rodea: Chris served as the project manager for this assignment. This included coordinating the team’s workflow, reducing the scope of the project accordingly to fit the timeline of the course, ensuring all other deliverables were completed by each deadline, and verifying all requirements of the project were completed. Chris was also responsible for the synthesizing and condensing the work that was done, into a cohesive report.

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